

PROFESSIONAL INTERNSHIP

PROJECT REPORT

College Admission Portal

Backend implementation and progress to date

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Progress report: completed backend work and remaining backend tasks

Report overview

This report records the college admission portal work implemented to date, with emphasis on backend services and college-managed content. It separates implemented scope from tasks that remain pending and does not claim that unverified tests have passed.

Executive summary

The project is a college discovery and admission-support platform with a Spring Boot REST backend. Work completed so far includes college registration and verification controls, student account and profile services, college discovery and recommendations, consent-based enquiries, college content management, media uploads, and role-scoped analytics. College content pages are stored per college, edited as drafts, and exposed publicly only after publication and college verification where applicable.

Recent CMS work expanded structured pages for departments, AICTE IDEA Lab, Research, IQAC, and the requested Campus Life pages: Campus Clubs, Professional Societies, Campus Celebrations, Campus Events, and Gallery. The public frontend is outside this backend implementation scope and is being handled separately.

Current implementation status

Backend capabilities and API contracts are in place in the repository. Maven compilation and automated tests remain unverified because the Maven Wrapper could not download its Maven distribution.

Project objectives

- Provide a shared college discovery and admission-support API for students and participating colleges.
- Let each college maintain its own profile, courses, departments, media, and structured public information.
- Separate college draft content from published content and protect college data through tenant-aware authorization.
- Provide student recommendations and enquiry workflows while limiting access to sensitive student activity.

1. Project scope and backend design

The backend is implemented as a Java 17+ Spring Boot application. REST controllers expose workflows to the independently developed client applications. Spring Data repositories persist relational records, while the college_content store holds configurable college-specific page data as structured JSON. Local development and integration tests are configured around H2; the production database is intended to be externally configured.

Core design principles

- College isolation: college administrators can manage only the college associated with their authenticated account.
- Verification before public listing: newly registered colleges begin pending and are not exposed in public college discovery until verified and active.
- Draft and publish workflow: editorial content can be saved privately and published separately.
- Public filtering: public retrieval excludes unpublished page content and inactive or unverified colleges.
- Image ownership: media is uploaded through the college-scoped media API; Campus Life pages reject media references owned by another college.
- Privacy by role: college-facing activity is aggregate-only; identifiable student-to-college visit details are restricted to platform administrators.

Main backend areas

- Authentication and authorization: account login, JWT-backed authenticated requests, college access checks, and role-aware admin endpoints.
- College management: signup, verification status, profiles, branding, courses, and public college listing/search.
- Student services: student profile and education data, preferences, saved/comparison workflows, recommendations, and enquiries.
- College CMS: About/Profile, departments, AICTE IDEA Lab, Research, IQAC, and Campus Life content.
- Media and reporting: image upload/serving, platform analytics, notifications, and generated report snapshots.

2. Implemented college and student workflows

College onboarding and discovery

- College signup creates a registered college account with a college-admin login. New registrations begin in PENDING verification state and are withheld from public lists.
- College login supports the configured email/login identifiers; passwords are stored as BCrypt hashes and are not returned in API responses.
- Platform administrators can review and update college verification status. Public lists and slug-based pages are restricted to active, registered, verified colleges.
- Public college discovery supports listing, search/filtering, profile retrieval, and course retrieval from the college-specific records/content.

Student services and admission support

- Student registration/profile services support education and marks records and saved preference data.
- Recommendation services match marks/percentage, course, district, and college type; configured course minimum-percentage cutoffs are applied where present.
- Students can manage saved colleges/comparisons and access their own relevant student activity through authenticated endpoints.
- Admission enquiries require authenticated student context and explicit consent. College access to contact information is limited to consented enquiries for the owning college.
- Enquiry status changes and other supported workflows can issue account-scoped notifications.

Analytics and privacy

Activity reporting is separated by audience. College-facing analytics are aggregate-only. A paginated student-to-college visit register, including identifiable visit details, is restricted to platform administrators. Student activity endpoints use the authenticated account rather than trusting a client-supplied student identifier.

3. College CMS and image management

The content-management APIs use college-scoped records and draft/publication status. Public clients receive only published sections/pages according to the college verification and activation state.

CMS sections implemented

- About/Profile: college profile, vision and mission, management profile, organizational structure, centers of excellence, accreditations, programmes, and placements as structured editable blocks.
- Departments: independent department directory and pages covering overview, vision/mission, courses, curriculum/regulations, labs, PEO/PO/PSO outcomes, HOD/faculty information, classrooms, and teaching resources. Department removal also removes its matching page data.
- AICTE IDEA Lab: lab overview, vision, mission/objective cards, grouped team profiles, and equipment details with images/links.
- Research: policy links, overview/year-breakdown tables, committees, yearly publication data, seed funding, research projects, facilities, and paginated public tables.
- IQAC: illustrated About section, function cards, member rows with public pagination, AQAR/minutes documents, and year-labelled reports/resources.
- Campus Life: configurable menu and page slugs; feature sections, clubs, professional societies, celebration schedules, event showcases with metric cards, and galleries.

College-owned image handling

College admins can upload JPEG, PNG, or WebP files up to 8 MB through POST `/api/colleges/{id}/media` using multipart field file. The response supplies a URL such as `/api/media/123`. Campus Life content supports these uploaded URLs for hero, feature, society, event, celebration, and gallery imagery; the CMS checks that referenced uploaded media belongs to the same college. Media list/delete and media serving endpoints are also available, with public serving gated by college verification.

Campus Life admin/public API

Admin: GET/PUT `/api/colleges/{id}/campus-life` and PATCH `/api/colleges/{id}/campus-life/publication`. Public: GET `/api/public/colleges/{slug}/campus-life` and GET `/api/public/colleges/{slug}/campus-life/pages/{pageSlug}`.

4. API workflow and publication lifecycle

A typical college content lifecycle begins after signup. A college admin authenticates, uploads any required images, saves the structured page as a draft, reviews it, then publishes it. Public consumers query the slug-based public endpoints; unverified, inactive, unpublished, or inactive page entries are not presented as published Campus Life content.

Selected endpoint groups

- College signup/login: POST /api/colleges/signup; POST /api/auth/login.
- Public college discovery: GET /api/colleges; GET /api/colleges/{slug}; GET /api/colleges/{slug}/courses.
- College content: GET/PUT /api/colleges/{id}/content/{section}; dedicated CMS routes are used for structured pages with validation/publication.
- Departments: /api/colleges/{id}/departments and /api/public/colleges/{slug}/departments/... routes.
- Image media: POST /api/colleges/{id}/media; GET /api/colleges/{id}/media/manage; GET /api/media/{mediaId}.
- Dedicated CMS pages: /idea-lab, /research-page, /iqac-page, and /campus-life under college admin/public route families.
- Admission workflow: recommendation preview, student recommendations, enquiries, notifications, and authenticated student portal endpoints.

Campus Life page content shape

A single college-scoped Campus Life configuration stores an ordered navigation menu and multiple page objects. Each page can include a title, intro paragraphs, optional hero/quote banner, and ordered sections. Image-bearing sections accept the college media URL returned by the upload endpoint. The public route returns the published menu/page payload for frontend rendering; no frontend rendering is claimed in this report.

Data and access handling

College-specific page data is persisted under the college content section key campusLifePage. Management controllers call the college access service before allowing reads/writes. Generic public content endpoints return the published Campus Life content rather than drafts. The detailed page endpoint resolves a page slug only within the published Campus Life record for the requested college.

5. Security, validation, and quality controls

- Passwords are hashed; JWT configuration expects an environment-provided secret rather than a known production fallback.
- College content writes require an authenticated user authorized for that college. Media list/upload/delete operations are also college-scoped.
- College signup and public disclosure are separate states; pending registrations do not appear as verified public colleges.
- Structured CMS payloads reject unsupported keys, bound text/list sizes, validate page slugs and display order, and restrict external links to HTTP(S) where applicable.
- Campus Life publication requires navigation/pages and suitable page copy; visual sections require uploaded/external images, and uploaded media references must belong to the same college.
- Generic public college/content responses substitute published content for CMS sections and do not expose drafts.
- Student-to-college visit details are platform-admin-only; college accounts receive aggregate analytics.

Testing and verification status

The repository includes Spring Boot integration tests using an isolated in-memory H2 database. Coverage in the project includes signup/default content, public verification filtering, recommendations, consent-based enquiries, college-scoped CMS workflows, draft/publication filtering, media handling, analytics privacy, and Campus Life page types including club/society lists, celebration schedules, event metrics, and galleries.

Important verification note

The Maven Wrapper was invoked, but it could not download Apache Maven 3.9.16 from repo.maven.apache.org. Therefore compilation and automated test results are not verified in this report. A successful git diff --check was observed; it is not a substitute for compilation or tests.

6. Backend work pending

The following items remain before the backend can be treated as fully verified and production-ready:

- Restore Maven distribution access or provide an installed Maven runtime, then run the complete backend test suite and resolve any compilation or integration failures.
- Run API-level verification with the actual frontend client for college signup, authentication, CMS draft editing/publication, image upload, and public page retrieval.
- Confirm the final Campus Life payload contract with the frontend developer, including navigation ordering, page layouts, event metrics, and image field usage.
- Complete production database configuration, schema migration strategy, backups, and data-retention review.
- Configure deployment-grade JWT secrets, CORS origins, HTTPS, request limits/rate limiting, upload storage/backup policy, and operational monitoring.
- Perform a final security/privacy review of tenant authorization, media access, public verification rules, and platform-admin-only identifiable activity reporting.

Work outside current backend scope

Frontend page construction and visual rendering are not included in this backend report; a separate developer is responsible for the frontend. The backend exposes published CMS content through documented REST endpoints for that client to render.

Conclusion

The project has established the core backend foundation for a multi-college admission and discovery portal and has implemented a broad set of student, college-admin, CMS, media, and platform-admin capabilities. The latest Campus Life CMS supports the specific requested college-managed pages and uploaded images without prefilled institution-specific content. The next priority is to regain Maven availability, run the full test suite, and verify the API contracts with the separately developed frontend before production deployment.

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